Student 2 · Master File

Everything you need in one place: the study, your Results part from zero, your slides, your questions.
Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad

> **Your role.** You present slides 13–23 (Results, about 6 minutes) and you are the team's statistics person. You own Chapter 3 (Results, 3.1–3.6), Tables 1–11 and Figures 2–8, and you must understand Section 2.5 (Statistical analysis). Handover line at the end of your part: “Those are our results. My colleague will now discuss what they mean.”

Contents

1. **The study in full (aim, hypotheses, methods, conclusions)**
2. **Whole-study essentials: two-minute script, core facts, terms**
3. **The Results as one story**
4. **Statistics from zero (with our numbers)**
5. **Every table and figure: why it is there and how to read it**
6. **Your chapter, section by section**
7. **Statistical analysis (Section 2.5, word for word)**
8. **Your slides: what to point at, and the full scripts**
9. **Questions and model answers, plus drills**
10. **Your key sources**
11. **Editing your chapter**
12. **Your 14-day plan, self-test and day-before checklist**
13. **Cheat sheet**

How to study this file

Day	Read
1	Section 1 (the study) and Section 2 (essentials). Say the two-minute script out loud three times.
2	Section 3 (the story) and Section 4, lessons 1–12.
3	Section 4, lessons 13–25. Explain each lesson out loud using our numbers.
4–5	Section 5 (every table and figure), with the thesis open beside you.
6	Sections 6 and 7 (your chapter and the methods).
7–8	Section 8: practise your slides alone, three times, with a timer.
9–10	Section 9: answer every question out loud; record yourself once.
11–14	Sections 10–12; rehearsals with the team; self-test until 100%; Section 13 in your pocket.

1 · The study in full

Title

Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad: An Online Cross-Sectional Survey

Abstract (as in the thesis)

Background: Many adults look up symptoms online before seeing a doctor, increasingly with artificial-intelligence (AI) chatbots. Health anxiety and cyberchondria (excessive, distressing online health searching) are thought to feed each other, but evidence from Iraq, and on AI chatbots, is scarce.

Objectives: To measure the association between health anxiety and cyberchondria among adults in Baghdad, and to describe online and AI health-information seeking and behaviour before consulting a doctor.

Methods: Cross-sectional online survey of 300 adults in Baghdad (July–September 2026) using the Short Health Anxiety Inventory (SHAI-14) and the Cyberchondria Severity Scale (CSS-12), analysed with FDR-corrected group comparisons, robust regression and bootstrapped mediation.

Results: Participants were mostly young (62.3% aged 20–29). AI chatbots were ticked by 78.0% (88.7% including those who named a chatbot), more than search engines (56.3%). Health anxiety correlated with cyberchondria ($r = 0.47$, 95% CI 0.38–0.55) and remained its strongest predictor after adjustment ($\beta = 0.48$), most strongly with excessive searching and post-search distress. Elevated health anxiety (SHAI-14 ≥ 18, a screening threshold, not a diagnosis) was found in 19.0%. After searching, 34.7% delayed or skipped care because they felt reassured and 31.7% never told their doctor, citing fear of judgement. The association between AI use and cyberchondria was not robust to how AI use was defined.

Conclusion: In this young, health-oriented convenience sample, health anxiety and cyberchondria were linked and AI chatbots were a major source of health information, without reliable evidence that they add to cyberchondria beyond health anxiety. Clinicians should ask patients, without judgement, what they have searched or asked an AI.

Keywords: cyberchondria; health anxiety; artificial intelligence; chatbots; Iraq.

Why the study was done (Rationale, 1.7)

This study addresses three gaps: it gives, to our knowledge, the first estimate of the association between health anxiety and cyberchondria (CSS-12) in Baghdad; it examines AI chatbots separately from search engines; and it links both constructs to what clinicians see before and during a consultation.

Aim, objectives and hypotheses (1.8, word for word)

Aim: To study the relationship between health anxiety and cyberchondria among adults in Baghdad who search for health information online, with a focus on AI chatbots and pre-consultation behaviour.

Objectives:

1. To describe where adults in Baghdad look for health information (online and through AI) and what they do before seeing a doctor.

2. To estimate levels of health anxiety and cyberchondria and the strength of their association, independently of socio-demographic and digital-use factors (H1, H2).

3. To examine statistical pathways consistent with a mutually reinforcing cycle (H3).

4. To examine how health anxiety and cyberchondria relate to pre-consultation behaviour, disclosure of searches and trust in physicians (H4).

Hypotheses: (H1) People with more health anxiety have more cyberchondria, especially on the distress and excessiveness dimensions. (H2) This holds after adjustment for socio-demographic and digital-use factors. (H3) Part of the link between health anxiety and post-search distress runs, statistically, through excessive searching. (H4) People with more cyberchondria make more urgent visits and multiple consultations and report a more strained doctor–patient relationship. H1 and H2 were primary and H3 and H4 secondary; all other analyses were exploratory (Figure 1).

Were the hypotheses supported? (memorise this table)

Hypothesis	Verdict	Evidence (where)
H1: health anxiety is positively associated with cyberchondria, most with distress and excessiveness	Supported	r = 0.47 (0.38–0.55); excessiveness 0.46 and distress 0.45 strongest (Table 5, Figure 4)
H2: independent of socio-demographic and digital-use factors	Supported	partial r = 0.48; health anxiety adds 18.4%, β 0.48 (Table 8)
H3: excessive searching mediates health anxiety → post-search distress	Supported statistically, not causally	indirect 0.16 (0.11–0.23), but reverse path also 0.15 (Table 11, Figure 7)
H4: cyberchondria ↔ care-seeking and strained relationship	Partially supported	only more disclosure survived FDR (AOR 0.57, q 0.012); urgent visits, strained relationship exploratory (Table 10)

> **Say:** *H1 and H2 were supported; H3 was supported statistically but not causally; H4 was only partially supported.*

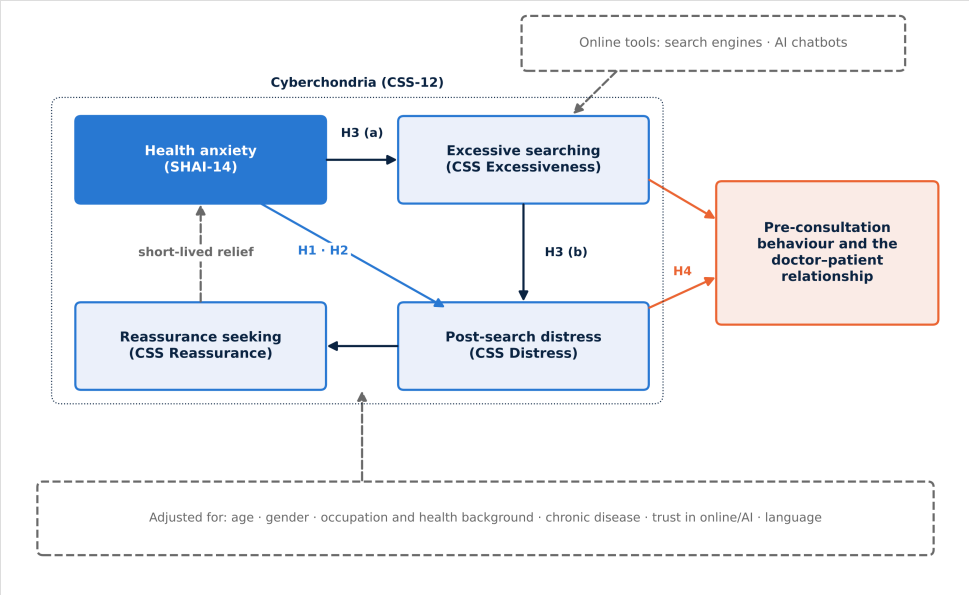


Figure 1: conceptual framework (where each hypothesis sits).

Methods (2.1–2.4, word for word; 2.5 is in Section 7)

This analytical cross-sectional study follows the STROBE [21] and CHERRIES [22] reporting guidelines. Data were collected online in Baghdad between 17 July 2026 and 11 September 2026 by convenience and snowball sampling through Baghdad-based social-media and messaging groups. Adults aged 18 or over who could read Arabic or English and gave informed consent were eligible. All completed submissions were analysed; no duplicate response patterns were found [Authors to state whether Google Forms limited submissions to one per account]. Three responses annotated by the team during data collection [authors to state the reason] were kept and tested in a sensitivity analysis.

2.2 Sample size

Using Fisher's z with two-sided α = 0.05 and 80% power, 194 participants are needed to detect a correlation of r = 0.20 [23]; meta-analytic estimates are far larger (r ≈ 0.62), so the target is conservative. For proportions, 300 participants give a margin of error of ±5.7%.

2.3 Questionnaire

A bilingual (Arabic–English) Google Form had five parts: (1) consent and socio-demographic data; (2) digital health-information seeking (tools, main AI chatbot, motives and trust in online/AI information); (3) the 14-item Short Health Anxiety Inventory (SHAI-14; items scored 0–3, total 0–42) [24], which has good reliability and validity [25,26]; (4) the CSS-12 (items scored 1–5, total 12–60; four 3-item subscales) [8]; and (5) behaviour after searching, disclosure of searches to the physician and the effect on the relationship with physicians. The Arabic wording was written by the team without formal cross-cultural adaptation [27]; published Arabic versions [9,28] were not used. The response options of one SHAI item (item 2) were displayed in a different order from the original and were scored by meaning.

2.4 Definitions

Elevated health anxiety: SHAI-14 ≥ 18, the cut-off that best separated people with health anxiety from healthy controls in a Swedish diagnostic study [26]; it is a screening threshold, not a diagnosis, so prevalence at 15, 22 and 29 is also reported.

High cyberchondria: CSS-12 > 33 (upper tertile), as the CSS-12 has no validated cut-off.

AI-chatbot user: ticked "AI chatbots" or named a chatbot; because 32 respondents named one without ticking it, all AI analyses were repeated with the strict (ticked-only) definition.

Health-related background: medical students, other health-science students and healthcare workers.

Ethics (2.7)

Participation was voluntary and anonymous; electronic informed consent (with confirmation of age ≥ 18) was required before any question, and no names or contact details were collected. [Ethical approval: committee name, approval number and date to be inserted.]

Conclusions (5.1, word for word)

1. AI chatbots, mainly ChatGPT, are a leading source of health information before consulting a doctor among digitally active adults in Baghdad (78.0% ticked; 88.7% including those who named one).

2. About one in five (19%) had elevated health anxiety on a screening threshold; younger people and reassurance seekers were most affected.

3. Health anxiety and cyberchondria are moderately and robustly associated (r = 0.47), compatible with a reinforcing cycle that this design cannot prove.

4. There is no reliable evidence that AI use adds to cyberchondria beyond health anxiety.

5. Searching shapes the consultation: about a third delay care when reassured and about a third never disclose their searches; higher cyberchondria goes with more disclosure.

Recommendations (5.2, word for word)

Clinicians: ask every patient, without judgement, what they have searched or asked an AI; treat repeated searching and reassurance seeking as possible signs of health anxiety; teach digital and AI health literacy in medical school.

Ministry of Health: the Ministry may consider Arabic digital and AI health-literacy campaigns that warn about false reassurance, and reliable Arabic triage sources.

AI developers: developers may consider anxiety-aware health answers that avoid escalating rare diagnoses and encourage appropriate consultation.

Research: longitudinal studies, probability samples and formally adapted Arabic scales; intervention trials are needed, as ClinicalTrials.gov lists only nine cyberchondria studies and none in Iraq [41].

Strengths and limitations (4.8, word for word)

Strengths include two established, reliable scales, formal Arabic–English invariance testing, separate assessment of AI chatbots, complete data, correction for multiple testing, robust and bootstrap methods, and sensitivity analyses. The main limitations are:

Design: cross-sectional data cannot establish direction; mediation describes statistical pathways only.

Sample: an online convenience sample, young and with many medical students (44.0%), so percentages cannot be generalised to Baghdad; residence was not verified by a question.

Measurement: Arabic wording without formal back-translation (although the CSS-12 was fully invariant); one SHAI item with reordered options; a poorly fitting one-factor SHAI-14 model; CSS item 3 mentions only search engines and may miss repeated chatbot use; behaviour, trust and disclosure measured with single, non-validated items; all data self-reported.

Analysis: many exploratory tests despite FDR correction; small subgroups (34 AI non-users); about five events per predictor in the logistic model; the AI result depends on how 32 respondents were classified.

2 · Whole-study essentials

> **The committee can ask any of you about any part.** You split the presentation, not the knowledge. Learn this part so well that you can explain the whole study without slides.

The study in two minutes (practise saying this out loud)

"Many people now look up their symptoms online before seeing a doctor, and more and more of them ask AI chatbots like ChatGPT. For some people this searching makes them more worried, not less. That is called cyberchondria, and it is closely linked to health anxiety. No one had studied this in Baghdad, and AI chatbots had hardly been studied next to search engines. We ran an online survey of 300 adults in Baghdad between 17 July 2026 and 11 September 2026, using two established scales: the SHAI-14 for health anxiety and the CSS-12 for cyberchondria. We found that AI chatbots are now the main source of health information: 78.0% ticked them, more than search engines (56.3%). About one in five (19%) had elevated health anxiety. Health anxiety and cyberchondria were moderately linked, r = 0.47, and this held after adjusting for everything else. The link ran mainly through excessive searching and post-search distress, which fits a vicious cycle, although one survey cannot prove direction. We found no reliable evidence that AI itself adds to cyberchondria beyond health anxiety. And searching shapes the consultation: about a third delay care because they feel reassured, and about a third never tell their doctor what they searched. So our main message for clinicians is simple: ask patients, without judgement, what they searched or asked an AI."

Core facts (memorise; examiners test these)

Topic	Fact
Design	Analytical cross-sectional online survey; STROBE and CHERRIES checklists
Who and when	**300** adults, Baghdad, 17 July 2026 – 11 September 2026; convenience + snowball sampling
Sample size	**194** needed for r = 0.20 (α 0.05, 80% power); margin of error ±5.7%
Sample profile	62.3% aged 20–29 · 56.3% female · **44.0%** medical students · 63.3% health background · 21.7% chronic condition · 70.7% answered in Arabic
Tools used	AI chatbots **78.0%** ticked (88.7% incl. named) · search engines 56.3% · medical platforms 18.3% · social media 10.7% · ChatGPT 59.3% · AI without any search engine 41.0%
Motives	convenience 80.0% · "do I need a doctor?" 57.7% · instant reassurance 51.0%
Health anxiety	SHAI mean 12.3 ± 5.8 (0–42) · **19%** elevated (≥ 18; 95% CI 15.0–23.8%) · 0.7% at ≥ 29
Cyberchondria	CSS mean 28.9 ± 8.4 (12–60) · excessiveness highest, compulsion lowest
Reliability	α 0.81 (SHAI), 0.86 (CSS); ω 0.82, 0.87 · CSS-12 4-factor CFA: CFI 0.95, RMSEA 0.070, SRMR 0.053
Arabic vs English	CSS-12 fully (scalar) invariant; SHAI-14 metric invariant; no item DIF after FDR
Main result	**r = 0.47** (95% CI 0.38–0.55); explains 22% of the variance · partial r = 0.48
By facet	excessiveness 0.46 · distress 0.45 · compulsion 0.25 · reassurance 0.24
Regression	R^2 0.030 (demographics) → 0.133 (+ digital use) → 0.317 (+ health anxiety, ΔR^2 18.4%); β = 0.48
Who is anxious	AOR **3.12** per SD of cyberchondria · 3.83 reassurance motive · 0.61 per older age group · AUC 0.83 → 0.79
Pathways	via excessive searching 0.16 (CI 0.11–0.23) · via distress 0.22 (direct → 0.01) · reverse 0.15
Behaviour	delay care when reassured **34.7%** · ignore symptom 30.0% · several doctors 25.3% · urgent care 22.0% · self-medication 21.3%

Doctor relationship	never tell the doctor **31.7%** · always tell 27.0% · trust strained 34.7% · higher cyberchondria → more disclosure (OR 0.57, q 0.012)
AI effect	+5.3 CSS points (inclusive) but +1.5, p = 0.31 (strict) → no reliable independent effect · 32 named-not-ticked · 34 non-users
Robustness	every sensitivity analysis left the main result unchanged (the sensitivity checks (Section 3.5))

Terms you must be able to explain in one sentence

Term	Plain explanation
Health anxiety	Persistent worry about having or developing a serious illness, kept going by misreading normal body sensations and by reassurance seeking.
Illness anxiety disorder	The DSM-5 diagnosis at the severe end of health anxiety. We did **not** diagnose it; we measured symptoms.
Cyberchondria	Excessive or repeated online health searching that increases rather than relieves distress.
SHAI-14	14 items, each scored 0–3 (total 0–42); higher = more health anxiety. ≥ 18 = elevated (screening threshold).
CSS-12	12 items scored 1–5 (total 12–60) with four 3-item facets: excessiveness, compulsion, distress, reassurance seeking.
Cross-sectional	Everything measured once, at one time point: shows associations, not cause and effect.
Convenience / snowball sampling	People who were easy to reach, who then shared the link with others. Quick, but not representative.
Mean ± SD	The average score and how spread out the scores are.
Median (IQR)	The middle score and the range of the middle half; used when data are skewed.
p-value	How surprising the result would be if there were truly no effect; $p < 0.05$ is called significant.
95% CI	The range in which the true value probably lies; narrower = more precise.
Correlation r	How closely two scores move together (0 = none, 1 = perfect). 0.47 is moderate.
Partial correlation	The same link after removing the effect of other variables.
Mann–Whitney / Kruskal–Wallis	Non-parametric tests comparing two / three or more groups, used because scores were not normally distributed.
χ^2 test	Tests whether two categorical variables are related (e.g., social-media use and elevated health anxiety).
FDR (q-value)	Correction for running many tests; a result is “robust” if $q < 0.05$, otherwise exploratory.
Linear regression, B and β	Predicts a score from several factors at once. B = change in CSS points per unit; β = the same on a standardised scale (to compare factors).
R^2 and ΔR^2	Share of variation explained by the model; ΔR^2 = extra share explained by adding a block of variables.
Logistic regression, AOR	Predicts a yes/no outcome. Adjusted odds ratio: how many times the odds change, holding other factors constant.
AUC	How well a model separates two groups (0.5 = chance, 1 = perfect).
Mediation	Splitting a link into a part that runs through a third variable (indirect) and the rest (direct). Statistical, not causal.
Moderation	Whether a third variable changes the strength of a link (e.g., does AI use make the link stronger? No).

Cronbach α / McDonald ω	Internal consistency: do the items of a scale hang together? ≥ 0.7 is acceptable, ≥ 0.8 good.
CFA (CFI, RMSEA, SRMR)	Checks the items group into the intended factors. Good: CFI ≈ 0.95+, RMSEA ≤ 0.08, SRMR ≤ 0.08.
Measurement invariance	Checks the scale measures the same thing in two groups (here Arabic vs English) before comparing them.
Sensitivity analysis	Re-running the main analysis in a different way to see if the result holds.

Never say (overclaims the committee will catch):

- "Health anxiety *causes* cyberchondria" → say "is associated with", "is compatible with a cycle".
- "19% *have* health anxiety disorder" → "19% scored in the elevated range on a screening tool".
- "AI makes people anxious" → "we found no reliable independent effect of AI use".
- "Our results represent Baghdad" → "a young, online, convenience sample; associations were consistent across subgroups".
- "The Arabic version is validated" → "we tested it: equal reliability and formal invariance, but no formal back-translation".

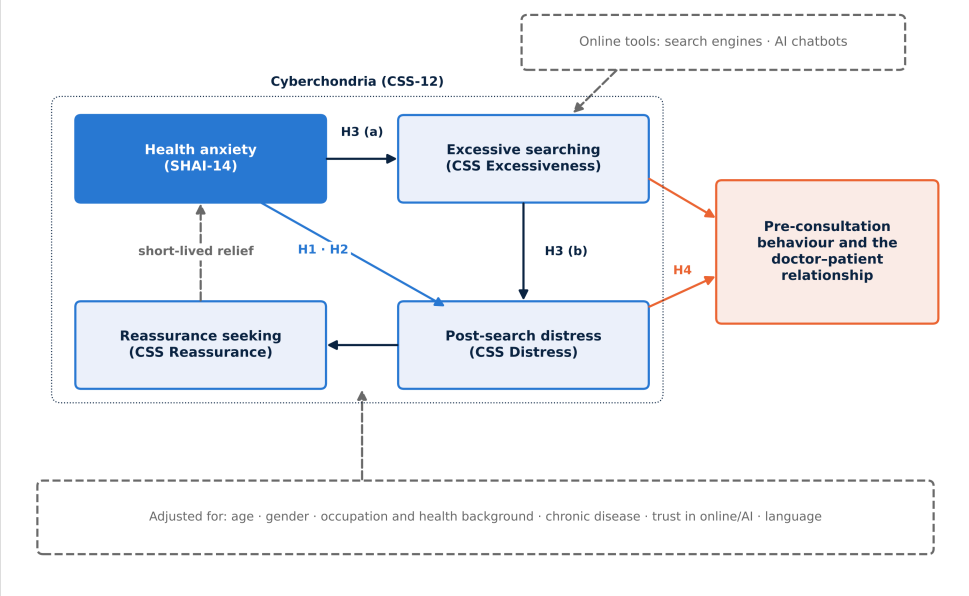


Figure 1 of the thesis: the conceptual framework. Learn to explain every arrow.

3 · The Results as one story

Every table and figure in Chapter 3 answers one question. They come in a logical order: first **who** answered, then **what they do**, then **how anxious** they are, then the **main link**, then **what else** could explain it, then **who is at risk**, then **behaviour**, then **pathways**, and finally **is it robust?**

Step	Question	Short answer	Where	Slide
1	Who took part?	300 young, educated adults; 44.0% medical students	Table 1	14
2	How do they search?	AI chatbots first (78.0%), search engines second (56.3%)	Table 2, Figure 2	15
3	What do they do after searching?	34.7% delay care when reassured; 31.7% never tell the doctor	Figure 3	22
4	How anxious are they? Do the scales work?	19% elevated; both scales reliable; CSS-12 same in Arabic and English	Tables 3 and 4	16
5	Are health anxiety and cyberchondria linked? (H1)	Yes, r = 0.47	Table 5, Figure 4	17, 19
6	Which groups score higher?	Chronic illness, reassurance seekers; AI users (but see step 7)	Tables 6 and 7, Figure 5	–
7	Does the link survive other factors? (H2) Who is at risk?	Yes; health anxiety adds 18.4%; AI effect not robust	Tables 8 and 9, Figure 6	18, 20, 23
8	Does it affect behaviour? (H4)	Cyberchondria → more disclosure (robust); others exploratory	Table 10	22
9	Is there a cycle? (H3) Is everything robust?	Compatible with a cycle, not proven; all checks unchanged	Table 11, Figures 7 and 8, the sensitivity checks (Section 3.5)	21

> **Say:** *Our results answer nine questions in order. In short: AI chatbots are the main source of health information, one in five has elevated health anxiety, and health anxiety and cyberchondria are clearly linked, r = 0.47, even after adjusting for everything else.*

4 · Statistics from zero (with our numbers)

Each lesson: the idea in plain words, our example, and what to say. Read them in order; each one builds on the one before.

1. Variables: two kinds

- **Categorical** = groups (gender, uses AI yes/no, tool profile). We describe them with **n and %**.
- **Continuous** = scores on a scale (SHAI 0–42, CSS 12–60). We describe them with **mean, SD, median**.

This one fact decides almost every choice of test and chart in the thesis.

2. Mean and SD

Mean = the average. **SD (standard deviation)** = how spread out people are around the average. About two-thirds of people lie within 1 SD of the mean.

```
SHAI mean 12.3 ± SD 5.8 → most people scored between about 6.5 and 18
```

> **Say:** *The average health-anxiety score was 12.3 out of 42.*

3. Median and IQR

The **median** is the middle person. The **IQR** is the range of the middle half. We use them when scores are skewed (not a symmetrical bell shape).

```
SHAI median 12 (IQR 8–16): half of people scored 8 to 16
```

4. Normal distribution and why it matters

Many tests assume scores form a bell curve. The **Shapiro–Wilk test** checks this. Our scores were slightly skewed ($p < 0.01$), so we used **non-parametric** tests (they do not need the bell shape) and **robust** regression.

> **Say:** *Our scores were not perfectly normal, so we used tests that do not assume normality.*

5. p-value

Imagine there is truly **no** link. The p-value is the chance of seeing a result as strong as ours just by luck. Small p (below 0.05) = "unlikely to be luck" = **significant**. It does **not** say how big or important the effect is.

```
r = 0.47, p < 0.001 → almost impossible to get this by chance if there were no
link
```

> **Trap:** p tells you if an effect exists, not how big it is. For size, look at r, β, OR.

6. 95% confidence interval (CI)

Our number is an estimate from 300 people. The CI is the range where the true value probably lies. Narrow CI = precise. If a CI for an odds ratio includes 1, the result is not significant.

```
r = 0.47, 95% CI 0.38–0.55 → the true correlation is probably between 0.38 and 0.55: always clearly above zero
```

7. Correlation r

How closely two scores move together: 0 = no link, 0.1 small, 0.3 moderate, 0.5 large, 1 = perfect. Positive = both go up together.

```
r = 0.47: people with more health anxiety tend to have more cyberchondria. r² = 22%: health anxiety explains about a fifth of the differences in cyberchondria.
```

> **Say:** *Health anxiety and cyberchondria rise together: r = 0.47, a moderate link.*

8. Pearson vs Spearman

Pearson r assumes a straight-line link; **Spearman ρ** uses ranks and does not need normal data. We report both; both were 0.47, so the result does not depend on the method.

9. Partial correlation

The same correlation after "removing" other variables (age, gender, background, chronic disease, trust, AI use). If it stays the same, those variables are not the explanation.

```
r = 0.47 → partial r = 0.48: unchanged
```

> **Say:** *The link stayed the same after adjusting for everything else, so it is not explained by age, gender or background.*

10. Comparing groups: Mann–Whitney and Kruskal–Wallis

To compare a **score** between groups without assuming normality: **Mann–Whitney U** for 2 groups, **Kruskal–Wallis H** for 3 or more.

```
Chronic condition: SHAI 14.2 vs 11.9 → Mann–Whitney p = 0.004
```

11. Chi-square (χ^2)

To test if two **categorical** variables are related, e.g. "elevated health anxiety yes/no" and "uses social media yes/no". It compares the counts we saw with the counts we would expect if there were no link.

12. Many tests → FDR correction (q-value)

If you run 162 tests at p < 0.05, about 8 will be "significant" by pure luck (162 × 0.05 ≈ 8). The **Benjamini–Hochberg FDR** correction adjusts p-values into **q-values**. q < 0.05 = **robust**; p < 0.05 but q ≥ 0.05 = **exploratory** (could be luck).

```
Never disclosing: CSS q = 0.012 (robust) · SHAI q = 0.098 (exploratory)
```

> **Say:** *Because we ran many tests, we corrected for multiple testing; results that did not survive are called exploratory.*

13. Linear regression, B and β

Regression predicts a score (CSS) from several factors at the same time, so each factor is "adjusted" for the others. **B** = how many CSS points change per 1 unit of the factor. **β** = the same on a common scale, so you can compare factors (bigger β = stronger).

```
SHAI: B = 0.68, β = 0.48. Someone with SHAI 20 vs 10 → about 6.8 CSS points
higher.
```

> **Say:** *Each extra point of health anxiety adds about 0.7 points of cyberchondria; it is by far the strongest factor.*

14. R^2, ΔR^2 and hierarchical regression

R^2 = share of differences the model explains (0–1). In **hierarchical** regression we add groups of variables in steps and see how much each step adds (**ΔR^2**).

```
Step 1 demographics R² 0.030 (3%) → step 2 digital use 0.133 (13%) → step 3
health anxiety 0.317 (32%). Health anxiety adds 18.4%.
```

> **Why this was chosen:** Hierarchical order shows that health anxiety matters even after everything else is already in the model.

15. Robust standard errors (HC3)

Regression assumes the "errors" are equally spread. Ours were not (Breusch–Pagan p = 0.004). HC3 robust standard errors fix the p-values and CIs so they are still valid. You only need: "we used robust standard errors because the residuals were uneven".

16. Odds and odds ratios (logistic regression)

Logistic regression predicts a yes/no outcome (elevated health anxiety yes/no). It gives **odds ratios (OR)**: OR 1 = no effect, above 1 = higher odds, below 1 = lower odds. **Adjusted** OR (AOR) = after accounting for the other factors.

```
Worked example from our data:
With elevated health anxiety: 52.6% had high cyberchondria → odds = 52.6/47.4 =
1.11
Without: 23.0% → odds = 23.0/77.0 = 0.30
Odds ratio = 1.11 / 0.30 ≈ 3.7
```

In the model: AOR **3.12** per **1 SD** of cyberchondria (1 SD ≈ 8.4 CSS points) means about three times the odds for every 8 extra points.

> **Trap:** AOR below 1 means protective. Age AOR 0.61: each older age group had lower odds.

17. AUC

How well the logistic model separates people with and without elevated health anxiety. Pick one person with and one without: AUC is the chance the model gives the higher risk to the right one. 0.5 = coin toss, 1 = perfect.

```
AUC 0.83, and 0.79 after correcting for optimism (bootstrap): good
```

> **Why this was chosen:** A model always looks better on the data it was built on, so we corrected the AUC with 500 bootstrap repeats.

18. Bootstrap

A computer trick: draw 5000 new samples of 300 people from our own data (with replacement), redo the analysis each time, and see how much the result varies. That gives a CI without assuming normality. We used it for mediation and to check the regression coefficient.

19. Mediation (indirect effect)

Does part of the link from X to Y go **through** a middle variable M? Path **a** = X → M; path **b** = M → Y (adjusted for X); **indirect** = a × b; **direct** (c′) = what is left.

```
Model A: health anxiety → excessive searching → distress
a = 0.46, b = 0.35 → indirect = 0.46 × 0.35 ≈ 0.16
total = 0.46, direct = 0.30 → about 35% goes through excessive searching
Model C: health anxiety → distress → reassurance seeking: indirect 0.22, direct
0.01 → almost all goes through distress
```

> **Trap:** Mediation in one survey is NOT proof of cause. The reverse order was also significant (0.15). Always say: compatible with a cycle, not proof.

20. Moderation (interaction)

Does a third variable change the **strength** of the link? E.g. is the SHAI–CSS link stronger in AI users? We tested 7 moderators; none were significant after FDR. So the link is similar in everyone.

21. Reliability: Cronbach α and McDonald ω

Do the items of a scale measure the same thing? 0.7 acceptable, 0.8 good, 0.9 excellent.

```
SHAI α 0.81, CSS α 0.86: good
```

22. CFA and fit indices

Confirmatory factor analysis checks whether the items group into the factors the scale claims (CSS-12 = four facets). Fit indices: **CFI** near 0.95 or higher = good; **RMSEA** and **SRMR** 0.08 or lower = acceptable.

```
CSS-12 four factors: CFI 0.95, RMSEA 0.070, SRMR 0.053 → acceptable · SHAI one
factor: CFI 0.83 → poor (it has two factors)
```

23. Measurement invariance

Before comparing Arabic and English answers, we must check the questionnaire works the same in both, like checking two rulers measure the same centimetre. Three steps: **configural** (same shape), **metric** (items equally strong), **scalar** (same starting points). If fit does not get worse at a step (CFI drop ≤ 0.01), that step passes.

CSS-12 passed all three. SHAI-14 passed the first two.

24. DIF

Differential item functioning: does a **single question** behave differently in Arabic vs English for people with the same total score? None did after FDR correction.

25. Sensitivity analysis

Re-doing the main analysis in another way (excluding doubtful responses, removing one item, another method) to see if the result changes. Ours did not change: r stayed 0.46–0.48.

5 · Every table and figure: why it is there and how to read it

The rules that decided every choice

If the data are...	...we used	Example
Exact numbers that people need to look up	a **table**	Table 1 participants, Table 8 regression
A pattern people should see at a glance	a **figure**	Figure 4 scatter plot
Percentages of groups (categorical)	**bar chart**	Figures 2 and 3
Two continuous scores together	**scatter plot**	Figure 4
Many correlations at once	**heatmap** (colour = strength)	Table 5 (heatmaps are only in the long version)
A score compared across groups, skewed data	**box plot** (median, spread, outliers)	Figure 5
Means of several facets in two groups	**bar chart with 95% CI**	Figure 8
Many odds ratios with CIs	**forest plot**	Figure 6
Paths between variables	**path diagram**	Figure 7

Also: the most important results get **both** a table (exact numbers) and a figure (the picture), e.g. the main correlation (Table 5 + Figure 4) and the logistic model (Table 9 + Figure 6). Less central results have only a table.

Table 1 · Who took part

> **Why this was chosen:** Every study must describe its sample first, so the reader knows who the results apply to. A table, because the reader needs exact n and %.

How to read it:

1. Rows = characteristics (age, gender, education, study/work, background, chronic disease, language).
2. Columns = n (number of people) and %.

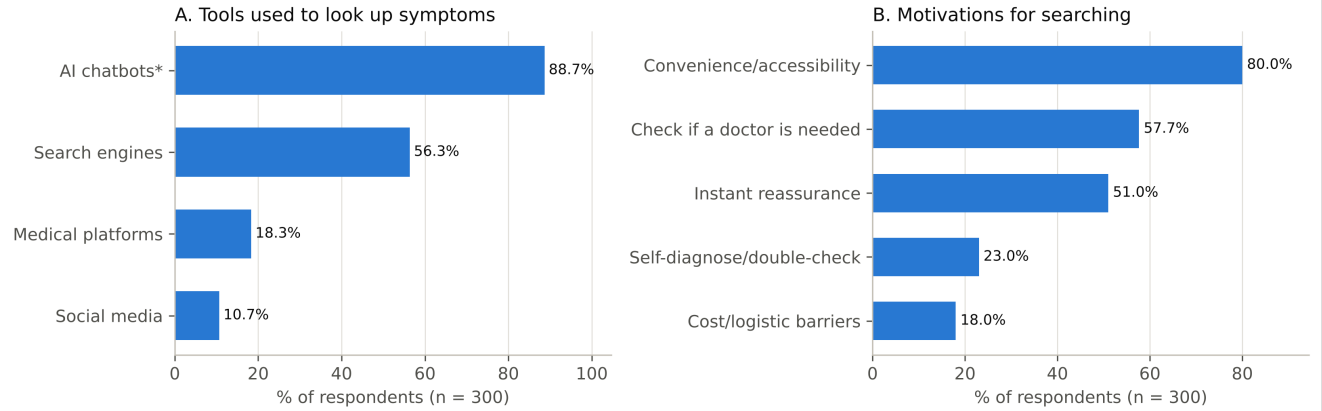
What it shows: A young sample (62.3% aged 20–29), 44.0% medical students, 63.3% with a health background.

> **Say:** *Our 300 participants were mostly young and educated; almost half were medical students. This matters when we interpret percentages.*

> **Trap:** Why so many medical students? Answer: the link spread through our own networks first; we say percentages cannot be generalised.

Table 2 + Figure 2 · How people search and why

> **Why this was chosen:** Tools and motives are categories with multiple answers, so percentages are best shown as bars (Figure 2); Table 2 keeps every exact count.



*Includes 32 respondents who named an AI chatbot without ticking the AI option. Multiple answers allowed.

How to read it:

1. Figure 2, panel A = tools; panel B = motives.
2. Each bar = % of all 300 who ticked that option.
3. Bars add up to more than 100% because people could tick several.

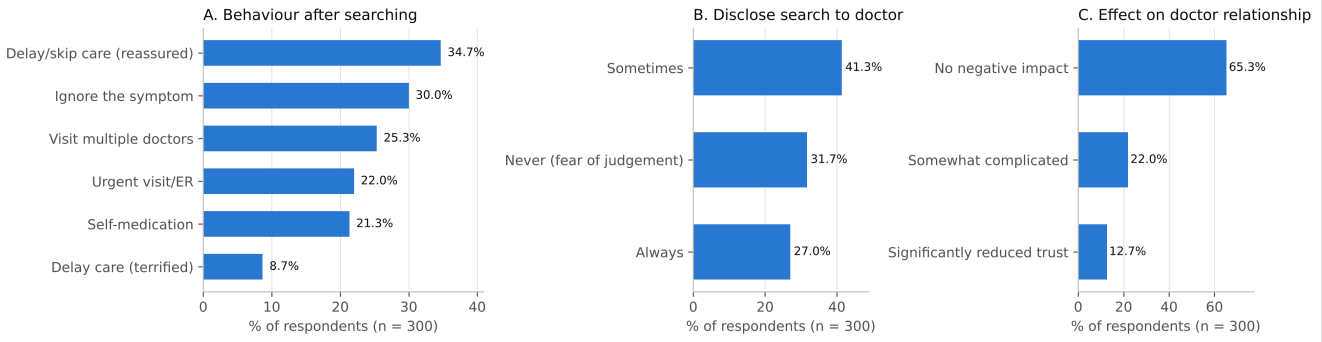
What it shows: AI chatbots 78.0% ticked (88.7% including those who named one) vs search engines 56.3%; convenience is the top motive.

> **Say:** *AI chatbots are now the main gateway: 78.0% ticked them, compared with 56.3% for search engines.*

> **Trap:** Why two AI numbers? 32 people named a chatbot but did not tick "AI"; we report both and test both.

Figure 3 · What people do after searching

> **Why this was chosen:** Three related sets of categories (behaviour, disclosure, relationship) shown side by side so the committee sees the whole consultation story in one picture.



How to read it:

1. Panel A: behaviours after searching (multiple answers).
2. Panel B: do they tell the doctor? (always / sometimes / never).
3. Panel C: effect on the relationship with doctors.

What it shows: 34.7% delay care when reassured; 31.7% never tell the doctor; 34.7% say trust was strained.

> **Say:** *Searching changes the consultation: one in three delays care because they feel reassured, and almost one in three never tells the doctor.*

Table 3 · Scores and reliability

Why this was chosen: Before using any score, we must show the scales work (reliability) and give their levels. A table because these are exact values.

How to read it:

1. Rows: SHAI, CSS and the four CSS facets.
2. Columns: mean ± SD, median (IQR), range, α (95% CI), ω.
3. The note under the table has the CFA fit.

What it shows: Both scales reliable (α 0.81 and 0.86); excessiveness is the highest facet, compulsion the lowest.

Say: *Both questionnaires were reliable. People search a lot, but it rarely disrupts their daily life.*

Table 4 · Prevalence at different cut-offs

Why this was chosen: The % with "elevated" health anxiety depends on the cut-off you choose. Showing four cut-offs is honest and stops the committee from accusing us of choosing the one that looks best.

How to read it:

1. Each row = a cut-off (15, 18, 22, 29).
2. Columns: n, %, 95% CI.
3. Bold row = 18, the one we use.

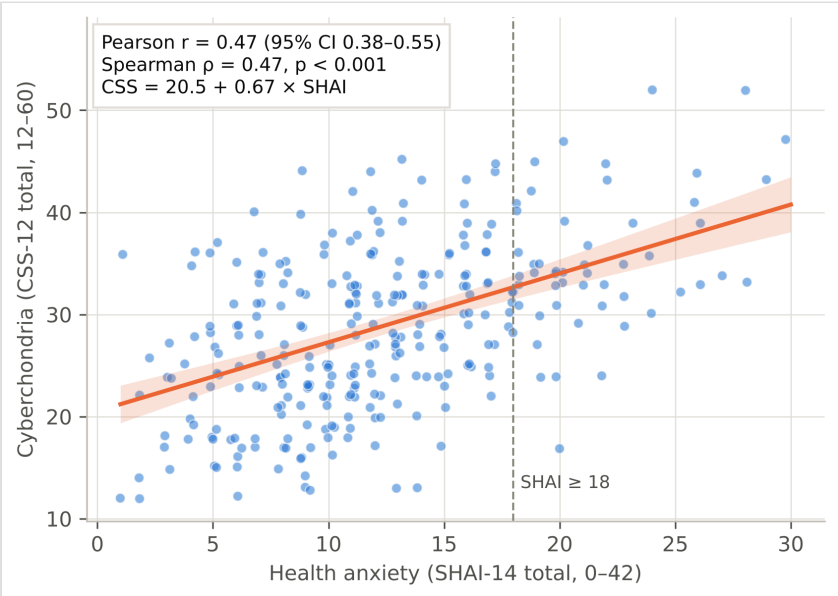
What it shows: 32.7% at 15, 19.0% at 18, 7.3% at 22, 0.7% at 29.

Say: *About one in five scored 18 or more; this is a screening threshold, not a diagnosis, and the table shows how the estimate changes with the cut-off.*

Trap: Why 18? It best separated people with health anxiety from healthy controls in the Swedish study.

Table 5 + Figure 4 · The main result

Why this was chosen: This answers our main question (H1), so it gets both: a figure to see the pattern, a table for exact numbers. A scatter plot is the standard picture for two continuous scores.



How to read it:

1. Each dot = one participant (x = SHAI, y = CSS).
2. The red line = best-fit line; the shaded band = its 95% CI.
3. The dotted vertical line = SHAI 18 cut-off.
4. Table 5: r for the total and for each facet, plus adjusted β.

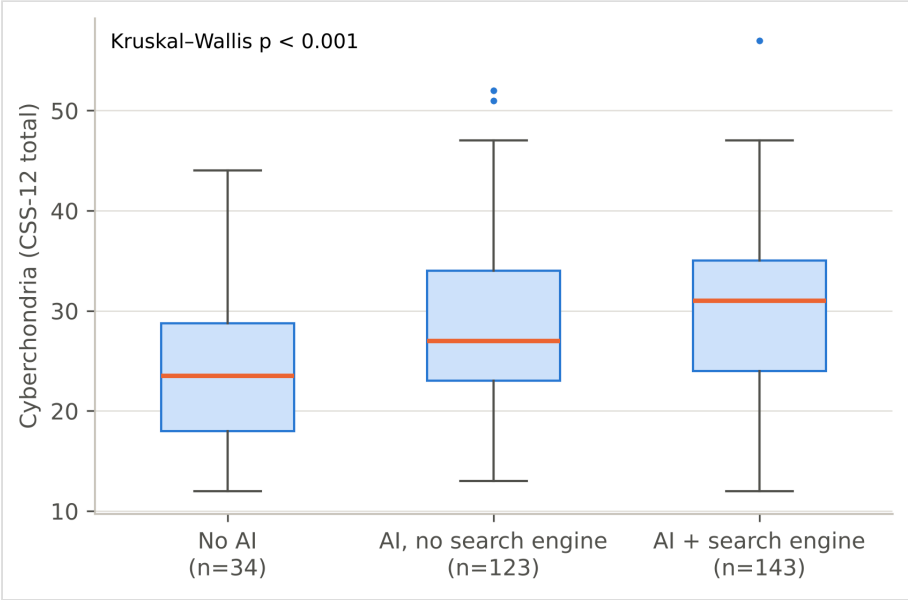
What it shows: Clear upward trend: r = 0.47; strongest for excessiveness (0.46) and distress (0.45).

> **Say:** *This is our main result: the more health anxiety, the more cyberchondria, r = 0.47, and it holds after adjustment.*

> **Trap:** Point at the line going up, then at the r value in the box. Do not say "causes".

Table 6 + Figure 5 · Which groups score higher

> **Why this was chosen:** Table 6 compares the scores across every participant group (exact means and p/q). Figure 5 is a box plot because CSS is skewed: it shows the median and spread, not just the mean.



How to read it:

1. Box plot: middle line = median; box = middle half of people; whiskers = range; dots = outliers.
2. Three groups: no AI, AI only, AI + search engine.

What it shows: Health anxiety higher with chronic illness (14.2 vs 11.9) and in reassurance seekers (13.8 vs 10.9); CSS higher in AI users (29.5 vs 24.0), highest in AI + search engine users.

> **Say:** *People who combine AI with search engines had the highest cyberchondria, but we will see this is not robust once we adjust.*

> **Trap:** Do not over-sell the AI difference; it disappears in the strict analysis.

Table 7 · Yes/no outcomes (χ^2)

> **Why this was chosen:** Some outcomes are yes/no (elevated health anxiety, high cyberchondria). χ^2 tests show which categories go with them. Only significant ones are listed to keep it short.

How to read it:

1. Each row: variable, % with the outcome in each group, Cramér's V (strength), p, q.

What it shows: With elevated health anxiety, 52.6% had high cyberchondria vs 23.0% without (OR ≈ 3.7).

> **Say:** *People with elevated health anxiety were much more likely to have high cyberchondria.*

Table 8 · What predicts cyberchondria (regression)

> **Why this was chosen:** Group comparisons look at one factor at a time; regression tests all factors together, so we know which matter independently (H2).

How to read it:

1. Rows = predictors; B = CSS points per unit; β = comparable strength; p.
2. Note under the table = R^2 for each step.

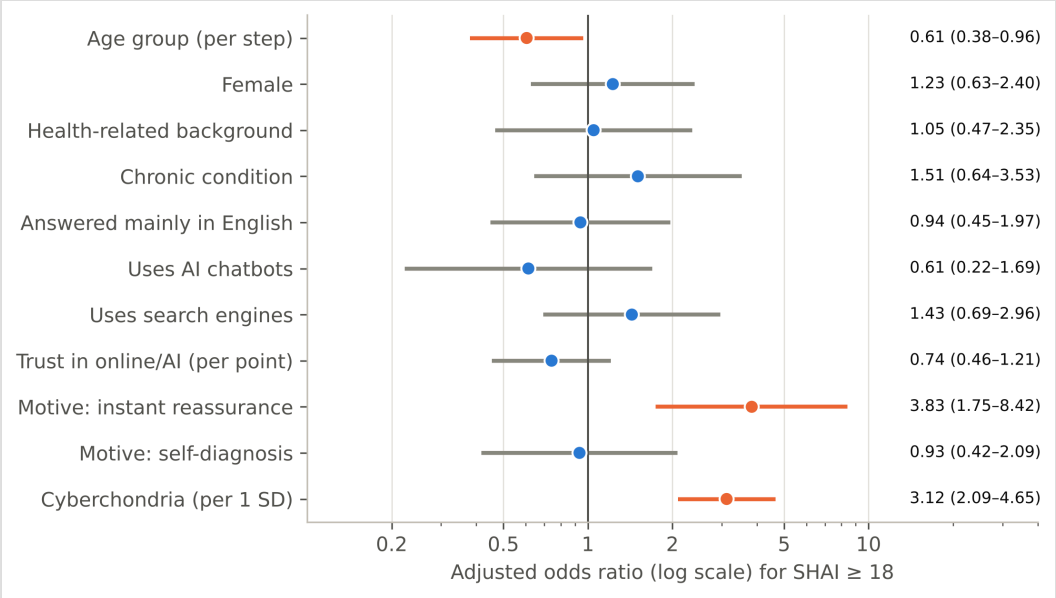
What it shows: Health anxiety β 0.48 (largest by far); R^2 rises from 0.030 to 0.317.

> **Say:** *Demographics explain only 3%; adding health anxiety explains a further 18.4%. Health anxiety is by far the strongest predictor.*

> **Trap:** AI row: significant here, but not with the strict definition (slide 23).

Table 9 + Figure 6 · Who has elevated health anxiety

> **Why this was chosen:** For a yes/no outcome we use logistic regression; a forest plot is the standard way to show many odds ratios with CIs at once.



How to read it:

1. Each dot = AOR; the line = its 95% CI.
2. Vertical line at 1 = no effect. If the line crosses 1 → not significant.
3. Log scale: 0.5 and 2 are the same distance from 1.
4. Orange = significant.

What it shows: Cyberchondria AOR 3.12, reassurance motive 3.83, age 0.61 (protective); AUC 0.83 → 0.79.

Say: *Three factors stand out: higher cyberchondria, searching for reassurance, and younger age.*

Trap: Only 57 events, so the reassurance odds ratio has a wide CI; say "exploratory".

Table 10 · Behaviour models

Why this was chosen: To test H4 we need adjusted models for each behaviour. A table because there are 8 outcomes × 2 scores: too many for a chart.

How to read it:

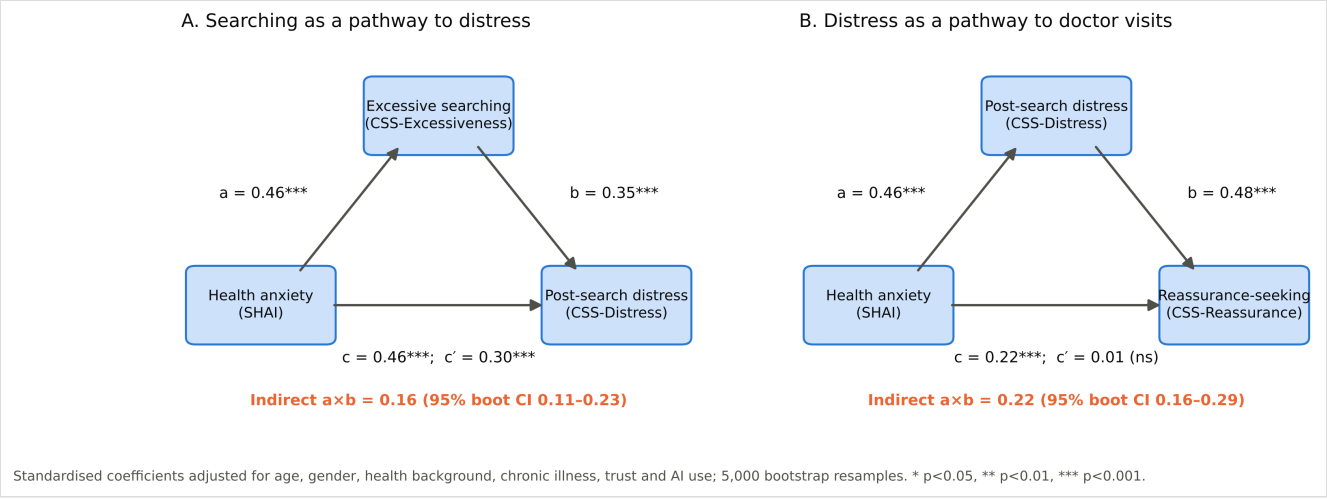
1. Each row = a behaviour; columns = AOR (95% CI) and p for SHAI and for CSS.

What it shows: Higher CSS → urgent visits (AOR 1.51), less ignoring (0.65), less never-disclosing (0.57, the only robust one).

Say: *Cyberchondria, not health anxiety, is what goes with behaviour; only the link with disclosure survived correction.*

Table 11 + Figure 7 · Pathways (mediation)

Why this was chosen: To test H3 (the cycle). A path diagram is the clearest way to show a, b and direct paths; the table adds the CIs and the reverse model.



Standardised coefficients adjusted for age, gender, health background, chronic illness, trust and AI use; 5,000 bootstrap resamples. * p<0.05, ** p<0.01, *** p<0.001.

How to read it:

1. Boxes = variables; arrows = paths with coefficients.
2. a = X → M, b = M → Y, c′ = direct.
3. Below each diagram: the indirect effect with its CI.

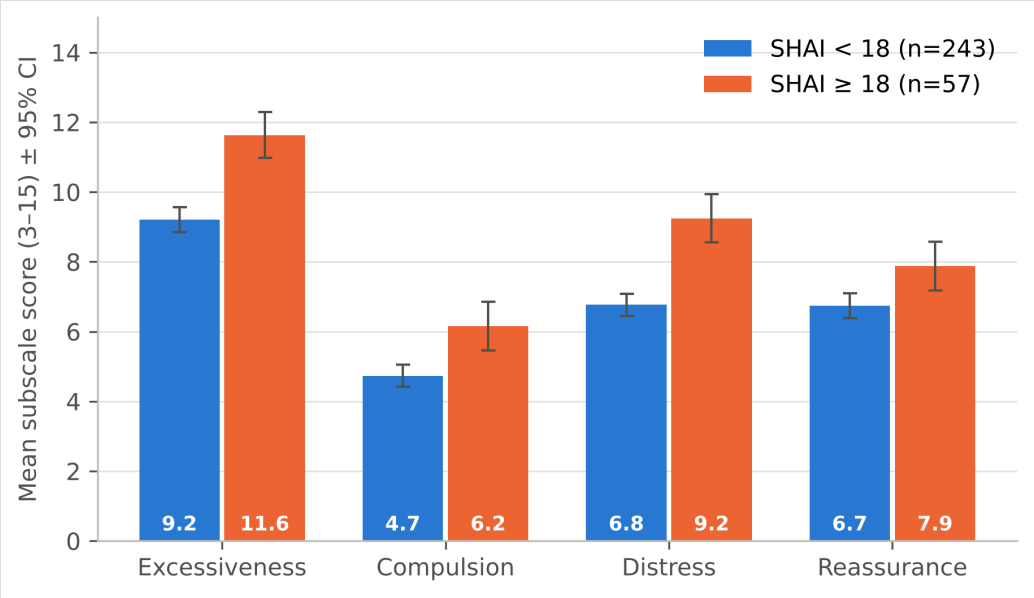
What it shows: Via excessive searching 0.16; via distress 0.22 (direct falls to 0.01); reverse also 0.15.

Say: *The pattern fits a cycle, but because the reverse direction also fits, we cannot say which comes first.*

Trap: Never say "proves".

Figure 8 · Facets by health-anxiety group

Why this was chosen: Shows simply that people with elevated health anxiety score higher on every facet, and which facets differ most. Bars with 95% CI are best for comparing means of two groups.



How to read it:

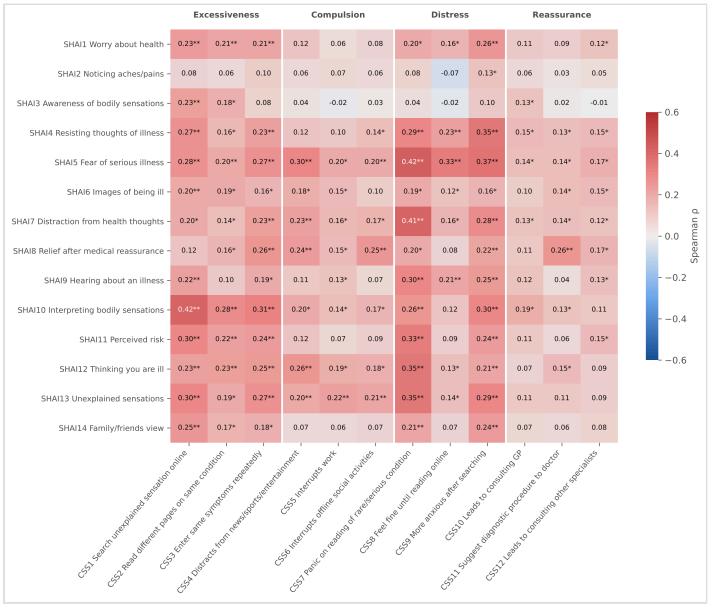
1. Blue = SHAI < 18 (243 people); orange = SHAI ≥ 18 (57 people).
2. Bar height = mean; small lines = 95% CI.

What it shows: Biggest gaps: excessiveness (9.2 vs 11.6) and distress (6.8 vs 9.2).

Say: *People with elevated health anxiety score higher on every part of cyberchondria, most of all on excessive searching and distress.*

Supplementary figure · Item by item (slide 19 only)

Why this was chosen: To see which exact questions drive the link (168 pairs). Only a heatmap can show 168 numbers readably, so it goes in the appendix.



How to read it:

1. Rows = 14 SHAI items; columns = 12 CSS items.

2. Colour = strength; stars = significant after FDR (**: also after the stricter Holm correction).

What it shows: 111/168 pairs significant; strongest: fear of serious illness ↔ panic on reading of a rare condition (0.43). Noticing aches is almost unrelated.

> **Say:** *What drives searching is fear and interpretation, not simply noticing symptoms.*

6 · Your chapter, section by section

For each section: its **purpose**, the **key points and numbers** (with the reference numbers used in the thesis), and what to **watch**.

3.1 Participants and online health-information seeking (Tables 1–2, Figure 2)

Who answered, how they search, and what they do after searching.

- 300 adults, no missing data; 62.3% aged 20–29; 56.3% female; medical students 44.0%; health background 63.3%; chronic condition 21.7%; Arabic 70.7%.
- AI ticked 78.0% (88.7% incl. named) vs search engines 56.3%; AI without any search engine 41.0%; ChatGPT 59.3%. AI use 96% at 18–19 y vs 72% at ≥ 40 y.
- Motives: convenience 80.0%, "do I need a doctor?" 57.7%, reassurance 51.0%.
- After searching: delay/skip care when reassured 34.7%; ignore 30.0%; several doctors 25.3%; urgent care 22.0%. Always tell the doctor 27.0%; never (fear of judgement) 31.7%; relationship strained 34.7%.

Watch: Message: a young, digitally active sample whose first "doctor" is often an AI; searching changes the consultation.

3.2 Levels of health anxiety and cyberchondria (Tables 3–4)

- SHAI 12.3 ± 5.8; elevated 19% (CI 15.0–23.8%); Table 4: 32.7% at ≥ 15, 19.0% at ≥ 18, 7.3% at ≥ 22, 0.7% at ≥ 29.
- CSS 28.9 ± 8.4; excessiveness highest, compulsion lowest.
- Reliability α 0.81 / 0.86; CSS-12 four-factor CFA acceptable (CFI 0.95, RMSEA 0.070).
- Arabic vs English (text only in the short version): CSS-12 fully invariant; SHAI-14 metric but not scalar invariant from a weak baseline, so SHAI language comparisons need caution.

Watch: Why several cut-offs? Because prevalence depends on the threshold; reporting all four is honest.

3.3 Main association (the invariance results (Section 3.2), Figure 3)

The central result (H1, H2).

- r = 0.47 (95% CI 0.38–0.55), ρ = 0.47, explains 22%; partial r = 0.48.
- Facets: excessiveness 0.46 and distress 0.45 stronger (Williams' T2, $p < 0.001$) than compulsion 0.25 and reassurance 0.24.
- Item level (text only): fear of serious illness and interpretation of sensations matter, mere awareness does not; 111/168 item pairs survive FDR.

Watch: H1 and H2 are answered here.

3.4 Factors associated (Tables 5–6, Figure 4)

162 group comparisons plus χ^2 tests.

- Health anxiety higher with chronic condition (q 0.028) and in reassurance seekers (q 0.002); age, education, occupation borderline (exploratory); gender no difference.
- AI users higher CSS (inclusive definition, q 0.005), highest in AI + search engine users (Figure 4); chatbot brand no difference.
- Elevated health anxiety → 52.6% vs 23.0% high cyberchondria (OR 3.71) (Table 6).

Watch: Know q-values: "survives correction" vs "exploratory".

3.5 Independent predictors (Tables 7–8, the correlation heatmap (long version))

- Regression: R^2 0.030 → 0.133 → 0.317; health anxiety adds 18.4%; B = 0.68, β = 0.48.
- AI: +5.3 inclusive vs +1.5, p 0.31 strict → no reliable independent effect.

- Logistic (57 events): AOR 3.12 cyberchondria per SD, 3.83 reassurance motive, 0.61 age; AUC 0.83 → 0.79; ≈ 5 events per variable → exploratory.
- Sensitivity (one sentence in the short version): r stayed 0.46–0.48 in every check.

Watch: You will be asked about AOR, AUC and why it was corrected.

3.6 Behaviour and pathways (Tables 9–10, Figures 5–6)

- Behaviour related more to cyberchondria than to health anxiety: urgent visits (AOR 1.51), less ignoring (0.65), strained relationship (ordinal OR 1.34).
- Only robust behavioural result: higher CSS → less likely to report never telling the doctor, i.e. more disclosure (AOR 0.57, q 0.012). Health anxiety → concealment AOR 1.47, q 0.098 (exploratory).
- Mediation: via excessive searching 0.16 (CI 0.11–0.23); via distress 0.22, direct 0.01; reverse 0.15. No moderator after FDR.

Watch: Always end with: “compatible with a cycle, not proof of direction”.

7 · Statistical analysis (Section 2.5, word for word)

You are the team's statistics person. Read this until you can explain each method in one sentence (Section 4 gives the plain-language version of each).

Analyses were scripted in Python 3 (two-sided $\alpha = 0.05$). Reliability was assessed with Cronbach's α and McDonald's ω [29], structure with confirmatory factor analysis (maximum likelihood), and equivalence of Arabic and English responders with multi-group measurement invariance [30] and item-level differential item functioning [31]. Associations were tested with Pearson and Spearman correlations (bootstrap CIs) and partial correlations; dependent correlations were compared with Williams' T2 test [32]. Scores were compared between groups with Mann–Whitney U or Kruskal–Wallis tests and categorical variables with χ^2 or Fisher's exact tests. Because many tests were run, p-values within each family were adjusted by the Benjamini–Hochberg false discovery rate (q-values) [33], and associations that did not survive correction are called exploratory.

Multivariable analyses comprised hierarchical robust (HC3) linear regression of CSS on socio-demographics, digital use and SHAI; logistic regression of elevated health anxiety (AUC corrected for optimism by bootstrap) and of each behaviour; and exploratory mediation models with 5000 bootstrap resamples [34], plus moderation by interaction terms. Sensitivity analyses excluded annotated and straight-line responses, removed SHAI item 2 and compared early with late responders.

8 · Your slides

A. What to point at and what to say (short version)

Short versions (three sentences each). The full scripts are in the speaker notes and in your Student 2 guide. Aim for 30–35 seconds per slide.

Slide	Point at	Say (short version)
13 · Results divider	Nothing to point at.	Thank you. I will present our results in six steps: who took part, how they search, the levels of health anxiety and cyberchondria, the main association, the predictors and pathways, and behaviour and AI.
14 · Participants (Table 1)	The highlighted rows: gender and medical students.	Table 1 shows our 300 participants. Most were young, 62.3% aged 20 to 29, and 44.0% were medical students. In summary, a young and educated sample, which we keep in mind when reading percentages.
15 · How people search (Figure 2)	The top bar of panel A (AI chatbots), then the four number boxes.	Figure 2 shows the tools people use. AI chatbots come first, 78.0%, ahead of search engines at 56.3%, and ChatGPT alone is used by 59.3%. In summary, for many people the first “doctor” is now an AI chatbot.
16 · Levels & reliability (Table 3)	The first two rows, then the 19% box.	Table 3 shows the scores. 19% scored in the elevated range of health anxiety, and both questionnaires were reliable, with alpha 0.81 and 0.86. In summary, our tools measured well, in both Arabic and English.
17 · Main result (Figure 4)	The rising line, then the r box.	This is our main result. Health anxiety and cyberchondria rise together, r = 0.47, and the link stays 0.48 after adjusting for everything else. In summary, hypotheses 1 and 2 are supported.
18 · What predicts cyberchondria (Table 8)	The R^2 rows, then the 18.4% box.	Table 8 is our regression. Demographics explain only 3% of cyberchondria, but health anxiety adds another 18.4%. In summary, how anxious you are matters far more than who you are.
19 · Item by item (supplementary figure)	The 0.43 and 0.42 boxes, then the “≈ 0” box.	This supplementary figure looks at every pair of questions. The strongest links are fear of serious illness and interpreting body sensations; simply noticing aches is almost unrelated. In summary, fear drives searching, not noticing.
20 · Who has elevated health anxiety (Figure 6)	The orange dots, then the four boxes.	Figure 6 shows the odds ratios. Higher cyberchondria triples the odds, searching for reassurance nearly quadruples them, and older age is protective; the model's AUC is 0.79 after correction. In summary, the person most at risk is young and searches for reassurance.
21 · Pathways (Figure 7)	Panel A indirect effect, then panel B direct effect, then the orange box.	Figure 7 shows the pathways. Part of the link runs through excessive searching (0.16), and almost all of the link to reassurance seeking runs through distress. In summary, this is compatible with a cycle, but one survey cannot prove the direction.
22 · Behaviour (Figure 3)	The first bar of panel A, then the four boxes.	Figure 3 shows what people do after searching: 34.7% delay care because they feel reassured, and 31.7% never tell their doctor. Higher cyberchondria goes with more disclosure, and that result survived correction. In summary, searching changes the consultation before it begins.
23 · AI chatbots	The table: inclusive vs strict rows.	Finally, AI. AI users scored 5.3 points higher, but with the strict definition the difference was only 1.5 and not significant, p = 0.31. In summary, there is no reliable independent effect of AI. I now hand over to my colleague.

B. Full scripts (speaker notes)

The script below is the speaker-notes text of your slides in the presentation. Practise until you can say it in your own words without reading.

Slide 13 · Results

Thank you. I will present the results in six steps: who took part, how they search, the levels of health anxiety and cyberchondria, the main association, the predictors and pathways, and finally behaviour and AI chatbots.

Slide 14 · Participants

Opening: Table 1 shows our 300 participants. Main finding: the sample is young, 62.3% aged 20 to 29, slightly more women at 56.3%, and 44.0% are medical students; 63.3% have a health-related background, and 21.7% reported a chronic condition. 70.7% answered mainly in Arabic. Closing: so this is a young, educated, digitally active sample, and that matters when we interpret percentages.

Slide 15 · How People Search

Opening: the study flowchart (long version) shows the tools people use and why. Main finding: AI chatbots are now the main gateway. 78.0% ticked AI chatbots, 88.7% if we include those who named one, compared with 56.3% for search engines. 41.0% use AI with no search engine at all, and ChatGPT alone is used by 59.3%. The top motives are convenience at 80.0%, checking whether a doctor is needed at 57.7%, and instant reassurance at 51.0%. Closing: in summary, for this sample, the first doctor is often an AI chatbot.

Slide 16 · Levels & Reliability

Opening: Table 3 shows the scores and reliability. Main finding: mean SHAI was 12.3 out of 42 and 19% scored 18 or more, our threshold for elevated health anxiety, with a 95% confidence interval of 15.0–23.8%. Mean CSS was 28.9 out of 60. Excessiveness was the highest facet and compulsion the lowest: people search a lot, but it rarely disrupts their lives. Both scales were reliable, alpha 0.81 and 0.86, and reliability was the same in Arabic and English responders. Formal invariance testing showed the CSS-12 works the same way in both languages, and no single item behaved differently. Closing: our tools measured well in this population.

Slide 17 · Health Anxiety ↔ Cyberchondria

Opening: this is our main result, Figure 3. Main finding: health anxiety and cyberchondria rise together, r = 0.47, 95% CI 0.38–0.55, so health anxiety explains about 22% of the variation in cyberchondria. After adjusting for every other variable, the partial correlation is still 0.48. The link is strongest for excessive searching, 0.46, and distress, 0.45, and weaker for compulsion and reassurance. Closing: in summary, hypotheses 1 and 2 are supported.

Slide 18 · What Predicts Cyberchondria?

Opening: Table 7 is the hierarchical regression for cyberchondria. Main finding: demographics alone explain only about 3% of cyberchondria. Adding digital-use variables brings it to 13%. Adding health anxiety adds another 18.4%, to about 32%. Health anxiety is the strongest predictor, standardised beta 0.48: each extra SHAI point adds about 0.68 CSS points. Closing: who you are matters little; how anxious you are matters most.

Slide 19 · Item by Item: What Drives Searching?

Opening: this supplementary figure breaks the association down item by item, all 168 pairs. Main finding: 111 pairs remained significant after FDR correction. The strongest links involve fear of serious illness and interpreting body sensations; for example fear of illness with panicking when reading about a rare condition. Simply noticing aches and pains was almost unrelated. Closing: what drives searching is fear and interpretation, not just noticing symptoms, exactly what the cognitive model predicts.

Slide 20 · Who Has Elevated Health Anxiety?

Opening: the correlation heatmap (long version) shows who has elevated health anxiety, in a multivariable logistic model. Main finding: three factors are independent. Each standard deviation of cyberchondria triples the odds, adjusted odds ratio 3.12. Searching for instant reassurance almost quadruples them, 3.83. Older age is protective, 0.61 per age step. The model discriminates well: AUC 0.83, 0.79 after correcting for optimism. Closing: the person most at risk is young and searches to feel reassured.

Slide 21 · Pathways: Is There a Cycle?

Opening: Figure 5 shows the pathways. Main finding: in model A, part of the link from health anxiety to post-search distress runs through excessive searching, indirect effect 0.16, CI 0.11–0.23. In model B, post-search distress carries almost all of the link from health anxiety to reassurance-seeking visits: the direct effect drops to 0.01. But the reverse direction also fits, 0.15. Closing: our data are compatible with the cycle, but one survey at one time point cannot prove direction. Hypothesis 3 is supported statistically, not causally.

Slide 22 · Behaviour Before the Consultation

Opening: Figure 2 shows what people do after searching. Main finding: the most common response is to delay or skip care because they feel reassured, 34.7%. Others escalate: urgent visits 22.0%, multiple doctors 25.3%. Only 27.0% always tell their doctor about their searches, and 31.7% never do, for fear of being judged. Higher cyberchondria was linked with more disclosure, odds ratio 0.57, and this survived FDR correction. 34.7% say searching complicated or reduced trust in their doctor. Closing: searching changes the consultation before it begins.

Slide 23 · AI Chatbots: An Independent Effect?

Opening: finally, did AI chatbots themselves matter? Main finding: AI users scored 5.3 points higher on cyberchondria after adjustment, mainly excessiveness and reassurance. But 32 people named a chatbot without ticking the AI option. When we use the strict definition, only people who ticked it, the effect disappears: B = 1.5, p = 0.31. And only 34 people were true non-users. AI use also did not change the strength of the anxiety–cyberchondria link. Closing: no reliable independent effect of AI. Absence of evidence, not evidence of absence. I now hand over to my colleague for the discussion.

> **Delivery for your part:** ≈ 6 minutes; about 25–35 seconds per content slide and 10 seconds per divider. Point at the number you are saying. Pause after each key number. End with: “Those are our results. My colleague will now discuss what they mean.”

9 · Questions and model answers

A. Questions on your part

Q: Why non-parametric tests?

Shapiro–Wilk showed the scores were not normally distributed; Mann–Whitney and Kruskal–Wallis do not assume normality. Welch tests confirmed the same conclusions.

Q: What is FDR and why did you use it?

With 162 comparisons, about 8 would be "significant" by chance at $p < 0.05$. Benjamini–Hochberg controls the proportion of false discoveries; results with $q \geq 0.05$ are labelled exploratory.

Q: Explain r = 0.47 to a non-statistician.

People with more health anxiety clearly tend to have more cyberchondria; anxiety accounts for about a fifth (22%) of the differences.

Q: Why both Pearson and Spearman?

Pearson for the linear correlation, Spearman as a check that does not depend on normality; both were 0.47.

Q: What does partial r = 0.48 tell us?

The link is not explained by age, gender, background, chronic disease, trust or AI use: it survives adjustment (H2).

Q: Why hierarchical regression?

To see what health anxiety adds after demographics and digital use: 18.4% extra variance, by far the biggest block.

Q: What do B and β mean in Table 8?

B = 0.68: each SHAI point adds about 0.7 CSS points. β = 0.48: the standardised effect, used to compare predictors.

Q: Why robust (HC3) standard errors?

The residuals were heteroskedastic (Breusch–Pagan $p = 0.004$); HC3 gives valid standard errors anyway.

Q: What is an adjusted odds ratio of 3.12?

For each standard deviation higher cyberchondria, the odds of elevated health anxiety are about three times higher, holding other factors constant.

Q: Why is the reassurance-motive AOR so wide?

Only 57 events for 11 predictors (≈ 5 per variable), so estimates are imprecise and probably optimistic; we label them exploratory.

Q: What is the AUC and why corrected?

How well the model separates elevated from non-elevated anxiety. Models look better on their own data, so 500 bootstrap refits corrected 0.83 to 0.79.

Q: Does the mediation prove a vicious cycle?

No. One time point, the reverse order fits too, and the facets come from one questionnaire. It is compatible with a cycle; only longitudinal data can show direction.

Q: Why does the AI effect disappear?

It depends on 32 people who named a chatbot without ticking it; they had the highest scores. Under the strict definition p = 0.31. Only 34 non-users made a small comparison group.

Q: Chronic illness was not significant in the logistic model. Why?

Dichotomising SHAI at 18 loses information and power. On the continuous score chronic illness is significant: p = 0.004 unadjusted and p = 0.029 adjusted (B +1.7).

Q: What do CFI, RMSEA and SRMR mean?

Fit indices for CFA. CSS-12 four factors: CFI 0.95, RMSEA 0.070, SRMR 0.053: acceptable. SHAI one-factor fitted poorly.

Q: Why does the SHAI fit poorly?

It is not strictly unidimensional; recent validations find two factors (illness likelihood and consequences). Reliability was still good (α 0.81).

Q: What is measurement invariance (the invariance results (Section 3.2))?

Testing in steps whether the scale works the same in Arabic and English: same structure, then same loadings, then same intercepts. CSS-12 passed all three; SHAI-14 passed the first two.

Q: Why is high cyberchondria defined as > 33?

There is no validated clinical cut-off for the CSS-12, so we used the upper tertile of our sample, as is common.

Q: Why is the prevalence so dependent on the cut-off (Table 4)?

Because most scores cluster just below the thresholds; that is why we report several cut-offs and say “elevated”, not “disorder”.

Q: What does the sensitivity checks (Section 3.5) show?

That the main correlation stays 0.46–0.48 whatever we exclude, and the regression is stable. Only the AI effect depended on a decision.

B. Questions anyone can be asked (know the answer even if it is not your part)

Q: Summarise your study in one minute.

Use the two-minute script in Part 1, shortened: design, sample, two scales, r = 0.47, AI result, behaviour, clinical message.

Q: What is your main finding?

Health anxiety and cyberchondria are moderately and robustly associated (r = 0.47), independent of other factors.

Q: What did you find about AI chatbots?

They are the main source (78.0% ticked), but there was no reliable independent effect on cyberchondria beyond health anxiety.

Q: What percentage had elevated health anxiety?

19% at SHAI ≥ 18, a screening threshold, not a diagnosis.

Q: What is your biggest limitation?

The cross-sectional design: no direction of cause.

Q: What is the clinical message?

Ask patients, without judgement, what they searched or asked an AI.

Q: Why did you split the presentation this way?

By chapter, but each of us studied the whole thesis.

Q: Were your hypotheses written before you collected the data?

Answer truthfully. A safe true answer: they follow from our objectives and previous research, and the questionnaire was designed to measure these things; we tested all of them and report every result, including those not supported.

Watch out: Never claim they were pre-registered.

Q: Who did the statistical analysis?

Answer honestly according to what you did. If AI tools helped with code or drafting, say so in line with your college policy and the declaration in the thesis.

C. Difficult or hostile questions

Q: "Your p-values are tiny because your sample is large."

The effect sizes are meaningful too: r = 0.47 is moderate and ΔR^2 18.4% is a medium effect ($f^2 \approx 0.27$).

Q: "You ran so many tests; something will be significant."

Exactly why we used FDR correction and separated robust from exploratory findings.

Q: "The mediators are subscales of the same scale. Isn't that circular?"

Partly, yes: shared method variance inflates the paths. That is why we call it an exploratory decomposition, not a mechanism.

D. Limitations and what was done about each (all students)

Limitation	What we did / how to answer
Cross-sectional design	Reverse-direction models; wording "compatible with"; longitudinal studies recommended
Convenience/snowball sample	Percentages not generalised; associations consistent across subgroups
Arabic wording (no back-translation)	Equal reliability and means; CSS-12 fully invariant, SHAI-14 metric; no item DIF
Single-item behaviour measures	Reported descriptively; associations labelled exploratory
Many tests	FDR correction; robust vs exploratory separated
Small subgroups (34 non-users)	Acknowledged; AI effect called "not robust"
Self-report and careless responding	Anonymous; straight-liners and annotated responses excluded in sensitivity analyses (no change)

> **How to answer any question:** pause → repeat the question briefly → answer in two or three sentences → stop. If you do not know: "We did not examine that directly; based on our data we would expect…, and it is a good question for future research." Never invent a number. If a teammate struggles: "If I may add…".

E. Drills: explain it simply

Cover the right column. Answer out loud in one or two sentences. Then check.

Explain…	Model answer
What is a p-value?	The chance of getting a result this strong by luck if there were really no effect. Below 0.05 we call it significant.

What does r = 0.47 mean?	When health anxiety goes up, cyberchondria tends to go up too; a moderate link, explaining about a fifth of the differences.
What is a confidence interval?	The range where the true value probably lies. Ours for r is 0.38 to 0.55.
Why did you correct for multiple tests?	With 162 tests, about 8 would be significant just by luck; FDR correction separates real results from probable luck.
What is regression for?	To see which factors matter when we look at all of them together.
What is β?	The strength of each factor on a common scale, so we can compare them. Health anxiety had the biggest, 0.48.
What is an odds ratio of 3?	The odds are three times higher. Below 1 means lower odds.
What is AUC?	How well our model tells anxious from non-anxious people apart: 0.79 is good.
What is mediation?	Checking whether part of a link passes through a middle step, here searching or distress.
Does mediation prove a cycle?	No. The reverse also fits, and it is one survey at one time point.
Why did the AI effect disappear?	It depended on 32 people who named a chatbot but did not tick the box; only 34 non-users. With the strict definition it was not significant.
Why non-parametric tests?	Our scores were not perfectly normal; these tests do not need that.
What is Cronbach's alpha?	Whether the questions of a scale measure the same thing; ours were 0.81 and 0.86, good.
What is measurement invariance?	Checking the questionnaire measures the same thing in Arabic and English, like checking two rulers. The CSS-12 passed fully.
What is a sensitivity analysis?	Doing the analysis in other ways to see if the result changes. It did not.
Why both a table and a figure for some results?	The figure shows the pattern; the table gives the exact numbers. We did both for the most important results.
Why a box plot for Figure 5?	Cyberchondria scores are skewed, so the median and spread are more honest than a bar of means.
Why a forest plot for Figure 6?	It shows many odds ratios and their CIs at once; anything crossing 1 is not significant.
Why is the item-by-item figure only on a slide, not in the thesis?	It has 168 numbers: the thesis reports the key result in one sentence; the full figure is in the long version.

10 · Your key sources

Every reference cited in your chapter, with its number in the thesis. Know the bold ones well: examiners often ask “who found that?”.

No.	Reference (as listed in Chapter 6)
[8]	McElroy E, Kearney M, Touhey J, Evans J, Cooke Y, Shevlin M. The CSS-12: development and validation of a short-form version of the cyberchondria severity scale. Cyberpsychol Behav Soc Netw. 2019;22(5):330-5. doi:10.1089/cyber.2018…
[9]	Hallit S, Rogoza R, Abi Semaan C, Azzi V, Sawma T, Obeid S. Validation of the Arabic version of the cyberchondria severity scale 12 items (CSS-12-Ar) among a sample of Lebanese adults. BMC Psychiatry. 2023;23(1):618. doi:10.1186/s…
[21]	von Elm E, Altman DG, Egger M, Pocock SJ, Gøtzsche PC, Vandenbroucke JP. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. Lancet. 2007;370…
[22]	Eysenbach G. Improving the quality of Web surveys: the Checklist for Reporting Results of Internet E-Surveys (CHERRIES). J Med Internet Res. 2004;6(3):e34. doi:10.2196/jmir.6.3.e34
[23]	Serdar CC, Cihan M, Yücel D, Serdar MA. Sample size, power and effect size revisited: simplified and practical approaches in pre-clinical, clinical and laboratory studies. Biochem Med (Zagreb). 2021;31(1):010502. doi:10.11613/BM.2…
[24]	Salkovskis PM, Rimes KA, Warwick HMC, Clark DM. The Health Anxiety Inventory: development and validation of scales for the measurement of health anxiety and hypochondriasis. Psychol Med. 2002;32(5):843-53. doi:10.1017/S00332917020…
[25]	Arnáez S, García-Soriano G, López-Santiago J, Belloch A. The Spanish validation of the Short Health Anxiety Inventory: psychometric properties and clinical utility. Int J Clin Health Psychol. 2019;19(3):251-60. doi:10.1016/j.ijchp…
[26]	Österman S, Axelsson E, Lindefors N, Hedman-Lagerlöf E, Hedman-Lagerlöf M, Kern D, et al. The 14-item short health anxiety inventory (SHAI-14) used as a screening tool: appropriate interpretation and diagnostic accuracy of the Swe…
[27]	Cruchinho P, López-Franco MD, Capelas ML, Almeida S, Bennett PM, Miranda da Silva M, et al. Translation, cross-cultural adaptation, and validation of measurement instruments: a practical guideline for novice researchers. J Multidi…
[28]	Alshayea AK. Latent structure, measurement invariance, and reliability of an Arabic version of the Short Health Anxiety Inventory. J Exp Psychopathol. 2020;11(2). doi:10.1177/2043808720912629
[29]	Hayes AF, Coutts JJ. Use omega rather than Cronbach's alpha for estimating reliability. But… Commun Methods Meas. 2020;14(1):1-24. doi:10.1080/19312458.2020.1718629
[30]	Leitgöb H, Seddig D, Asparouhov T, Behr D, Davidov E, De Roover K, et al. Measurement invariance in the social sciences: historical development, methodological challenges, state of the art, and future perspectives. Soc Sci Res. 20…
[31]	Ćosić Pilepić A, Mohorić T, Takšić V. Multidimensional polytomous DIF detection methods: a Monte Carlo simulation study. Appl Psychol Meas. 2026;50(7):01466216261446320. doi:10.1177/01466216261446320
[32]	Steiger JH. Tests for comparing elements of a correlation matrix. Psychol Bull. 1980;87(2):245-51. doi:10.1037/0033-2909.87.2.245
[33]	Benjamini Y, Hochberg Y. Controlling the false discovery rate: a practical and powerful approach to multiple testing. J R Stat Soc Series B Methodol. 1995;57(1):289-300. doi:10.1111/j.2517-6161.1995.tb02031.x

[34]	Hayes AF. Introduction to mediation, moderation, and conditional process analysis: a regression-based approach. 3rd ed. New York: Guilford Press; 2022.

11 · Editing your chapter

> **The Word file is the master copy.** Edit only the .docx, in one shared OneDrive / Word Online file with **Review → Track Changes** on. Keep an untouched copy (Thesis_ORIGINAL.docx). Avoid Google Docs: it breaks the automatic numbering.

For your chapter

- Results is mostly numbers: **do not change any number**, p, q, CI, test name or table reference.
- You may reword the connecting sentences in your own words; keep one main message per paragraph.
- Never type table or figure numbers by hand in captions; in text, keep "Table n / Figure n" exactly.
- If a number or sentence looks wrong, add a comment and ask; do not "fix" statistics yourself.
- Read 2.5 Statistical analysis until you can explain each method in one sentence; you will be asked.

Never touch

- Numbers, %, p, q, CI, r, OR, β, n
- Reference numbers like [18] (move them with their sentence)
- The reference list
- Table/figure captions, table of contents and lists (automatic fields)
- Heading numbers and styles
- Times New Roman 14 and page numbering

The rewrite method (paragraph by paragraph)

1. Read until you understand it; look up anything unclear.
2. Close it and explain it out loud.
3. Write your version from that explanation.
4. Compare: same facts, numbers and [n]; nothing claimed more strongly.
5. Read aloud; change any phrase you would never say.

After editing

1. Ctrl + A, then F9, to update the table of contents, the lists and the captions.
2. Search for "[" to find placeholders left.
3. Swap chapters with a teammate for checking.
4. Export to PDF and look at every page.

> **About AI use, honestly.** Check your college's AI policy with your supervisor and declare AI assistance if required. A sentence you can adapt: *"The statistical analysis code and an initial draft of this report were prepared with the assistance of an AI tool (Claude, Anthropic). All analyses were checked, and the text was reviewed and rewritten by the authors, who take full responsibility for its content."* Do not use paraphrasing tools or tricks to fool detectors: they make the text worse and can count as misconduct. Real ownership (your words, your understanding, your edit history) is the best protection.

12 · Plan, self-test and checklist

14-day plan

Day	Task
1	Read the Abstract, 1.8 and Chapter 5. Learn the two-minute script (Part 1).
2	Read the whole thesis once, marking what you cannot explain.
3	Core facts and terms (Part 1). Self-test on core facts.
4	Your chapter, sections in Part 2 (first half).
5	Your chapter, second half; your key sources (Part 3).
6	Start rewriting your chapter (Part 6).
7	Finish rewriting. Self-test (Part 8).
8	Swap chapters with a teammate; check accuracy and voice.
9	Your slides: read the scripts (Part 4), then present them alone 3 times.
10	Questions A–D out loud (Part 5). Record yourself once.
11	Team: full rehearsal with timing and handovers.
12	Team: mock committee; each student asks the others their Part 5 questions.
13	Fix weak answers; second full rehearsal; final Word checks.
14	Day before: checklist (last page); one calm run-through; sleep.

Self-test

Cover the answers, answer out loud, then check. Repeat until all are correct.

#	Question	Answer
1	Main correlation with 95% CI?	r = 0.47 (0.38–0.55)
2	Partial correlation?	0.48
3	Which two facets correlate most with health anxiety?	Excessiveness (0.46) and distress (0.45)
4	R^2 of the three regression blocks?	0.030 → 0.133 → 0.317
5	β and B for health anxiety in the final model?	β 0.48, B 0.68
6	Three independent predictors of elevated health anxiety and their AORs?	Cyberchondria 3.12, reassurance motive 3.83, age 0.61
7	AUC apparent and corrected?	0.83 → 0.79
8	Indirect effect via excessive searching?	0.16 (CI 0.11–0.23)
9	Direct effect of HA on reassurance seeking after distress?	0.01 (not significant)
10	The only behavioural association that survived FDR?	Higher cyberchondria → less never-disclosure (AOR 0.57, q 0.012)
11	AI effect inclusive vs strict?	+5.3 vs +1.5 (p 0.31)
12	Prevalence at cut-offs 15/18/22/29?	32.7% / 19.0% / 7.3% / 0.7%
13	Which scale is fully invariant across languages?	CSS-12 (scalar); SHAI-14 metric only
14	How many of 168 item pairs survived FDR?	111

15	Reliability α of both scales?	0.81 and 0.86
16	Main correlation (everyone)?	r = 0.47
17	Elevated health anxiety (everyone)?	19%
18	AI chatbots ticked (everyone)?	78.0%
19	Never tell the doctor (everyone)?	31.7%
20	Sample size (everyone)?	300

Day-before checklist

✓	Item
☐	All placeholders filled (names, supervisor, college, logo, degree, year, ethics, Google Forms question, Acknowledgment)
☐	Fields updated (Ctrl + A, F9); PDF checked page by page
☐	AI declaration added if required
☐	Presentation: names on slide 1, ethics number on slide 10; USB + PDF + email copies; clicker tested
☐	I can give the two-minute summary and my slides (13–23) without notes
☐	I have answered every question in Part 5 out loud
☐	I scored 100% on the self-test
☐	Printed thesis copy if required; arrive early; agree who stands where

Generated from the study outputs and the presentation, so every number and slide script here matches the thesis and the deck.

13 · Cheat sheet

What	Number	Where
Participants	300; 62.3% aged 20–29; 44.0% medical students	Table 1
AI chatbots / search engines	78.0% (88.7% incl. named) / 56.3%; ChatGPT 59.3%	Figure 2
Delay care when reassured / never tell the doctor	34.7% / 31.7%	Figure 3
SHAI mean / CSS mean	12.3 / 28.9	Table 3
Reliability α	0.81 / 0.86	Table 3
Elevated health anxiety	19% (CI 15.0–23.8%); 32.7 / 19.0 / 7.3 / 0.7% at 15/18/22/29	Table 4
Invariance	CSS-12 scalar ✓; SHAI-14 metric ✓, scalar ×; no DIF	the invariance results (Section 3.2)
Main correlation	r = 0.47 (0.38–0.55); partial 0.48; r^2 22%	Table 5, Figure 4
Facets	excess 0.46, distress 0.45, compulsion 0.25, reassurance 0.24	Table 5
Regression R^2	0.030 → 0.133 → 0.317; ΔR^2 18.4%; β 0.48; B 0.68	Table 8
Logistic	AOR 3.12 / 3.83 / 0.61; AUC 0.83 → 0.79	Table 9, Figure 6
Behaviour (robust)	CSS → never disclose AOR 0.57, q 0.012	Table 10
Mediation	via excess 0.16 (0.11–0.23); via distress 0.22, direct 0.01; reverse 0.15	Table 11, Figure 7
AI effect	+5.3 inclusive vs +1.5 strict (p 0.31); 32 named-not-ticked; 34 non-users	Table 8, slide 23
Item level	111/168 FDR; top ρ 0.43	the item-by-item figure (long version)
Robustness	r 0.46–0.48 in every check	the sensitivity checks (Section 3.5)

> **Say:** *Health anxiety and cyberchondria travel together, r = 0.47; AI chatbots are the main source but have no reliable independent effect; and searching changes the consultation before it begins.*